Samuel Pereira

Music Theory · Musical Analysis · Computational Musicology

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I'm a researcher in music theory and computational musicology developing the Fourier Qualia Space, a novel framework applying discrete Fourier transforms to model harmonic-quality motion and ambiguity across four centuries of Western art music, with a special focus on post-tonal repertoires. My work combines quantitative analysis with music-theoretical enquiry, bridging computational methods and humanistic interpretation to understand how harmonic qualities convey hierarchical and syntactic organisation.

ACADEMIC AND RESEARCH POSITIONS

Invited Assistant Professor *(Professor Auxiliar Convidado)* — 2026–

University of Coimbra, Faculty of Arts and Humanities
Department of History, European Studies, Archaeology and Arts

– Scientific area: Musical Studies. Fixed-term contract in public functions, part-time (20 %), 09/2026–02/2027, renewable under Article 293 of the LTFP.
– Teaching in the master's programmes in *Computer Music and Sound Design* and *Art Studies*.

Seconded Researcher — 2024

Marie Skłodowska-Curie Actions Staff Exchanges "EA-DIGIFOLK", grant agreement 101086338

– Project: *A European and Ibero-American Approach for the Digital Collection, Analysis and Dissemination of Folk Music* (2023–2026), coordinated by the University of Salamanca, with partners in Portugal (INESC TEC and the University of Coimbra), Spain and Ireland.
– Seconded from INESC TEC for two periods at the University of Salamanca: 24 March–16 April and 17 August–15 September 2024.
– Designed a survey and interviews with professors of the Salamanca Conservatory to assess and quantify how musicians perceive similarity in melody, harmony, rhythm, form, texture and text.

Doctoral Researcher

2021–2025

INESC TEC, Sound and Music Computing Lab

Faculty of Engineering, University of Porto (FEUP)

– Held under FCT doctoral studentship 2021.05059.BD, full time, 1 October 2021–30 September 2025.

– Conducted doctoral research on symbolic harmonic analysis using discrete Fourier transforms. Developed computational tools for large-scale corpus analysis and harmonic-quality visualisation, with outputs including peer-reviewed publications on musical syntax and open-source software.

EDUCATION

DEGREES

PhD in Digital Media

2021–2026

University of Porto, Faculty of Engineering (FEUP)

– Thesis: *The Fourier Qualia Space: Interaction, Ambiguity, and Hierarchy in Music Harmony*

– Supervisors: Prof. Gilberto Bernardes, Prof. José Oliveira Martins

– Public defence: 15 June 2026. Approved by unanimity.

– Examination committee: Dr. António Coelho, Dr. Martin Rohrmeier (École Polytechnique Fédérale de Lausanne), Dr. Matt Chiu (Baldwin Wallace University), Dr. Isabel Pires (NOVA University Lisbon), Dr. Gilberto Bernardes, and Dr. António Sá Pinto

MSc in Multimedia

2017–2020

University of Porto, Faculty of Engineering (FEUP)

– Dissertation: *A Sintaxe Harmónica Atonal no Espaço de Fourier: Para Uma Representação Esférica das Fases*

– Specialisation in interactive music and sound design

– Supervisor: Prof. Gilberto Bernardes

– Final grade: 18/20

BA in Musical Composition

2014–2017

Escola Superior de Música e Artes do Espectáculo (ESMAE), Polytechnic of Porto

– Final grade: 17/20

CERTIFICATIONS

Certificado de Competências Pedagógicas (CCP) — 2021

Escola de Negócios e Administração (ENA)

– Professional teaching certification for vocational training

– Final grade: 5/5

GRANTS AND AWARDS

Doctoral Studentship — 2021–2025

Fundação para a Ciência e a Tecnologia (FCT)

– Reference: 2021.05059.BD

– Title: *Cartografias Harmónicas e a Conceptualização do Movimento Musical: Uma Sintaxe para a Música Atonal no Espaço de Fourier*

– 48 months of funding awarded in a national competitive call, 1 October 2021–30 September 2025, hosted by FEUP and INESC TEC (Sound and Music Computing Lab).

– Doctoral research in computational musicology, investigating whether post-tonal music possesses a coherent syntax in the unfolding of harmonic qualities (e.g., chromaticity, octatonicity, and diatonicity) over time.

PUBLICATIONS

JOURNAL ARTICLES

Pereira, Samuel, Gilberto Bernardes, and José Oliveira Martins. *"Qualia Motion in Fourier Space: Formalizing Linear, Nondirected and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1"* — 2026

Music Theory Spectrum 48 (1): 126–51. Published online 17 December 2025.

Bernardes, Gilberto, Nádia Carvalho, and Samuel Pereira. *"FluidHarmony: Defining an Equal-Tempered and Hierarchical Harmonic Lexicon in the Fourier Space"* — 2022

Journal of New Music Research 51 (2–3): 142–61. Open access.

UNDER REVIEW

Pereira, Samuel, Gilberto Bernardes, and José Oliveira Martins. *"Statistically Modelling Qualia Syntax Through Fourier Space: A Three-role Probabilistic Framework Where Ambiguity Becomes Structure"* 2026

Manuscript submitted to **Musicae Scientiae** (SAGE), August 2026; awaiting peer review.

CHAPTERS IN PEER-REVIEWED PROCEEDINGS

Pereira, Samuel, Giovanni Affatato, Gilberto Bernardes, and Fabian C. Moss. *"Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity"* 2024

In **Mathematics and Computation in Music: 9th International Conference, MCM 2024**, ed. Thomas Noll et al., 317–29. Lecture Notes in Computer Science 14639. Cham: Springer.

Pereira, Samuel, and Gilberto Bernardes. *"The Fourier Phase Sphere: A Method for Computer-Assisted Atonal Composition"* 2022

In **Anais do V Congresso Internacional de Música e Matemática: Homenagem a Jamary Oliveira**, 40–54. Rio de Janeiro: Universidade Federal do Rio de Janeiro, Escola de Música. ISSN 2594-9128. Presented online in 2020.

DOCTORAL THESIS

Pereira, Samuel. *"The Fourier Qualia Space: Interaction, Ambiguity, and Hierarchy in Music Harmony"* 2026

PhD diss., **University of Porto**. Open access from 14 December 2026.

SOFTWARE AND DATA

Affatato, Giovanni, Samuel Pereira, Gilberto Bernardes, and Fabian C. Moss. *Qualia module of the midiVERTO web application* 2023

Web tool of the Digital and Cognitive Musicology Lab, **École Polytechnique Fédérale de Lausanne**. Qualia trajectories and wavescapes through the Fourier Qualia Space.

Pereira, Samuel. *sml-pereira/phd-fqs* 2026

Zenodo. Code and data of the doctoral thesis, all versions. Code under MIT; data under CC BY 4.0.

Pereira, Samuel. *sml-pereira/qualia-syntax* 2026

Zenodo. Code, data and robustness tests of the manuscript under review, all versions. Code under MIT; data under CC BY 4.0.

CONFERENCE PRESENTATIONS

"Tonfeld Theory and the Discrete Fourier Transform: A Bridge Between Hermeneutic and Mathematical Accounts of Extended Tonality" 2026

EnIM 26: XV Encontro de Investigação em Música (SPIM), Royal Building of Mafra, Mafra, 4–6 November.

"The Fourier Qualia Space: Interaction, Ambiguity, and Hierarchy in Music Harmony" 2026

EuroMAC 11: 11th European Music Analysis Conference, Escola Superior de Música de Catalunya (ESMUC), Barcelona, 7–10 September.

"Qualia Motion in Fourier Space: Formalising Linear, Nondirected and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1" *(invited talk)* 2025

CEIS20 Colloquium "Music as Interdiscipline", University of Coimbra, Coimbra, 24 September.

"Historical Evolution of Harmonic Qualities in Music: Unravelling Fourier's Tonality, Entropy, and Ambiguity from the 16th to the 20th Centuries" 2024

Digital Technologies Applied to Music Research: Methodologies, Projects and Challenges, NOVA FCSH, Lisbon, 27–29 June.

"Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity" 2024

MCM 2024: 9th International Congress on Mathematics and Computation in Music, University of Coimbra, Coimbra, 18–21 June. With G. Affatato, G. Bernardes and F. C. Moss.

"A Propósito da Representação Geométrica do Movimento Musical: o Tonnetz e os Espaços de Fourier" 2022

EnIM 22: XI Encontro de Investigação em Música (SPIM), Department of Communication and Art, University of Aveiro, Aveiro, 10–12 November.

"The Atonal Harmonic Syntax in Fourier's Space" 2021

EnIM 21: X Encontro de Investigação em Música (SPIM), University of Coimbra, Coimbra, 11–13 November. With Gilberto Bernardes.

"Harmonic Cartographies and the Conceptualisation of Musical Movement: A Syntax for Atonal Music in Fourier Space" 2021

DMDS 2021/22: Digital Media Doctoral Symposium, University of Porto, Porto, 9 November.

"The Fourier Phase Sphere: A Method for Computer-Assisted Atonal Composition" 2020

MusMat 2020: 5th International Conference on Music and Mathematics, online.

EDITORIAL AND PEER REVIEW

Reviews Editor *(Editor de Recensões)* 2026–

Revista Portuguesa de Musicologia, nova série · ISSN 2183-8410

– Since September 2026. Commissioning, editing and scheduling reviews of books, CDs, concerts, digital musicology corpora, and software.

– Published in partnership by SPIM (Sociedade Portuguesa de Investigação em Música), INET-md (Instituto de Etnomusicologia — Centro de Estudos em Música e Dança), and CESEM (Centre for Music Studies).

RESEARCH UNITS AND MEMBERSHIPS

Member 2021–

INESC TEC, Centre for Telecommunications and Multimedia · Sound and Music Computing Lab

Member 2021–

Sociedade Portuguesa de Investigação em Música (SPIM)

TECHNICAL SKILLS

Programming	Python · LaTeX · Shell scripting · Git
Packages	`NumPy` · `SciPy` · `Pandas` · `Matplotlib` · `scikit-learn` · `Seaborn` · `tqdm` · `music21`
Quantitative	Discrete Fourier transform analysis · PCA · *t*-SNE · Hierarchical clustering · Markov chain analysis · *n*-gram modelling · Entropy metrics · Zipf's law
Analytical	Pitch-class set theory · Transformational and neo-Riemannian theory · Schenkerian analysis
Corpora	Symbolic encodings (MusicXML, MIDI) · Large symbolic corpora · Automated analysis pipelines · Feature extraction · Open publication of code and data (GitHub, Zenodo)
Visualisation	Publication-quality figures · RadViz projections · *Wavescapes* · Dendrograms · Heatmaps · Interactive plots
Tools	Jupyter Notebook · VS Code · Overleaf · Tableau · Jamovi · RAWGraphs · Office Suite · Sibelius · MuseScore · Pure Data · Max/MSP · Reaper

LANGUAGES

Portuguese *native* · English *C1* · Spanish *reading proficiency*

OTHER ACTIVITIES

European Youth Parliament 2005–2006

– National session delegate, Covilhã, Portugal.

– International session delegate, Paris, France.